the environment is generally explained in the form of a "**Material Balance Model**" developed by AlenKneese and R.V. Ayres. The model considers the total economic process as a physically balanced flow between inputs and outputs. Inputs are bestowed with physical property of energy which is received from the environment. The interdependence of economics and environment is given in the figure10.1 and flow diagram

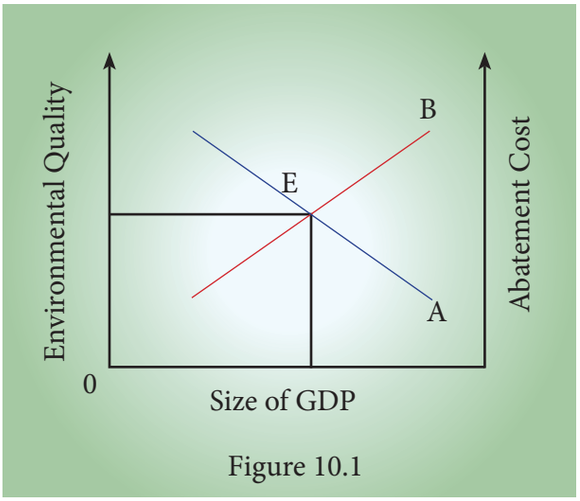
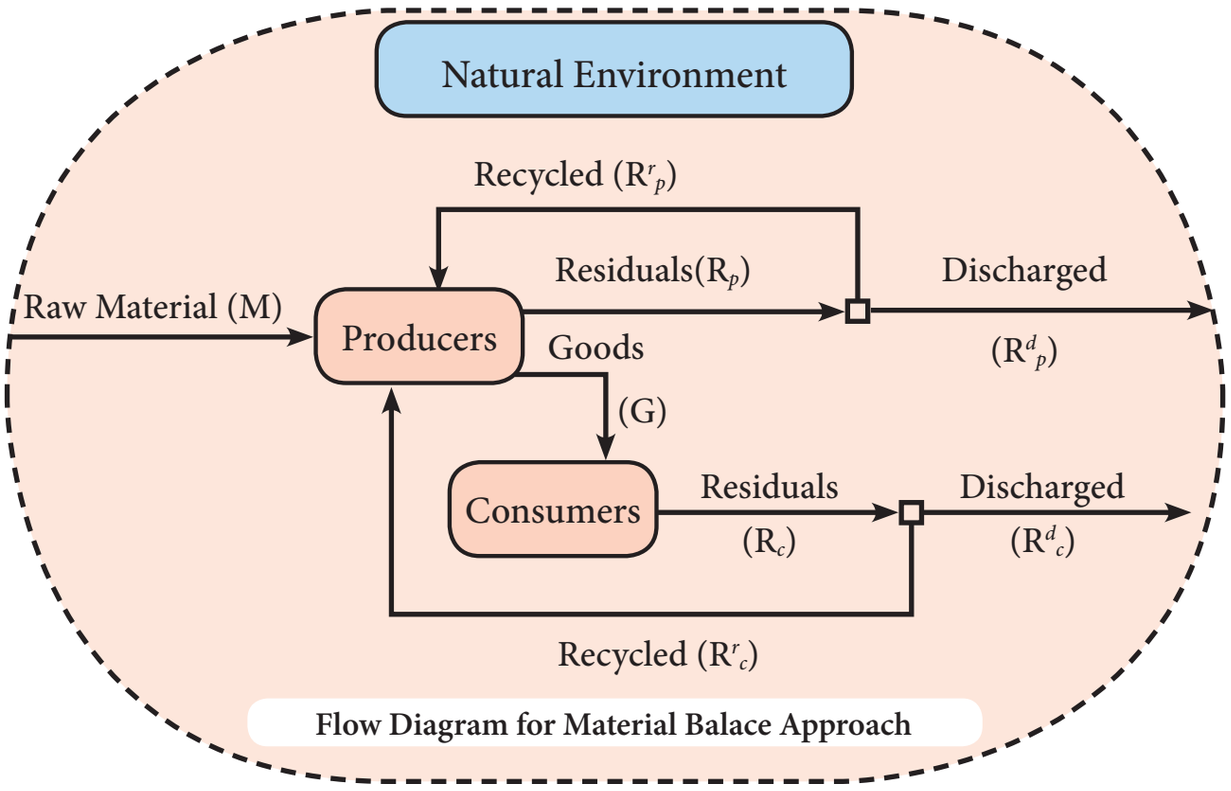


Figure 10.1



Flow Diagram for Material Balace Approach

The first law of thermodynamics, i.e. the law of conservation of matter and energy, emphasizes that in any production system "what goes in must come out". This is known as the Material Balance Approach or Material Balance Principle. The material flow diagram implies that mass inputs must equal mass outputs for every process. Moreover, all resources extracted from the environment eventually become unwanted wastes and pollutants. Production of output by firms from inputs resulting in discharge of solid, liquid and gaseous wastes. Similarly, waste results from consumption activities by households. In short, material and energy are drawn from environment, used for production and consumption activities and returned back to the environment as wastes. In its simple form the Material Balance Approach can be put in form equation.

12-Economics-Chapter_10.indd 214 05-02-2020 13:04:41